

WEEK 04 / HISTORICAL FOUNDATIONS

Portable Gaming, Mobile Gaming, and Monetization

Mobile gaming, portable devices, and the economics of monetization.

This week we talk about mobile gaming, portable devices, and the economics of monetization. We will discuss how mobile platforms have transformed the industry by introducing new business models, gameplay experiences, and accessibility for players on the go.

When we use the term "portable gaming" in this course, we are referring to 3 distinct but closely related things:

- Single-game handheld units, such as those released by Mattel and Tiger in the 80s and 90s
- Handheld game consoles that can play multiple video games, such as the Nintendo Game Boy, Nintendo Switch, and Steam Deck
- Mobile gaming involving apps played on smart phones

Why are we including monetization this week? Mobile gaming has either benefitted from or fueled the fire of the free to play model and the various monetization strategies for free to play games, so this seems like a good time to discuss it.

18 RESOURCES / 1 TOPIC**TOPIC 01**

Portable and Mobile Gaming

REQUIRED

The Evolution of Handheld Gaming

YOUTUBE | ~25 MIN

https://www.youtube.com/watch?v=cj5bNIXk17Q&ab_channel=Boo

Some highlights:

- Texas Instruments and Unisonic had calculators that could play games
- Mattel and Rockwell International released their first handheld electronic games in 1976 and was selling 500k units a week at one point
- Coleco (The Connecticut Leather Company) jumped into the handheld market, starting the handheld console wars
- Several other manufacturers started building similar devices
- Nintendo and Sharp released Game & Watch handheld consoles in 1980 after Gunpei Yokoi came up with the idea after watching someone play with a calculator on a train and they produced 50 different consoles over

10 years, selling millions of units

- The D-Pad that is still used today was introduced on the Donkey Kong Game & Watch
- Nintendo Game Boy launched in 1989, the first handheld console that could play different games using cartridges, sold 64.4 million units for \$90 each (BTW, Tetris was bundled with the units)
- Several competitors came out, but were more expensive, lower quality, and didn't sell as well
- Game Boy Color came out in 1998
- Today, Game Boy series is third best selling console of all time
- Bandai (now Bandai Namco, which owns Pac-Man, Tekken, Dark Souls, Elden Ring) released a competitor that became the second highest selling handheld console in Japan at the time
- PlayStation Portable released in 2004 right after Nintendo release the DS, both marvels and sold well, kicking off serious competition between Sony and Nintendo in the handheld market, but Nintendo won with the 3DS
- Nintendo continues to dominate the market, now with Nintendo Switch 2
- Valve Steam Deck allows people to play desktop PC games on a handheld, and there are other devices that could be thought of as competitors, but Nintendo is far ahead of them

LO 1: Students will be able to explain the historical development of video games, from early arcade games to contemporary digital gaming platforms.

REQUIRED

Buying Tiger Handheld LCD Games in The 90s \- We Were Duped ! | Retro Mysteries

[YOUTUBE](#) | ~9 MIN

https://www.youtube.com/watch?v=xsmsS5ejb2Q&ab_channel=GenerationGapGaming

Highlights:

- Tiger Electronics released a number of handheld games in the 1990s, although they started in the 1980s
- They focused on licensing popular intellectual property
- Retailed for around \$20
- Games were simple but fun at the time

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REQUIRED

The History, Evolution, and Future of Mobile Gaming

[ARTICLE](#) | ~20 MIN

<https://gameopedia.com/blogs/the-history-evolution-and-future-of-mobile-gaming>

Highlights:

- Mobile gaming projected to be \$139.5 billion industry by 2026
- Tetris was the first game released on a mobile phone, but Snake on a Nokia phone in 1997 was when mobile phone games started to get traction
- In 2001, Space Invaders, an arcade great, was released for mobile phones
- Apple released iPhone in 2007 and launched AppStore in 2008, which started the modern mobile game revolution
- Angry Birds came out in 2009 and when they ported it to Android, they released both paid and ad-supported free versions
- Farmville on Facebook had over 80 million players by February 2010
- Candy Crush Saga released in October 2012, also featuring Facebook integration, advertising, and in-app purchases, earning revenues of \$1.88 billion by the end of 2013
- Clash of Clans was earning \$5.15 million per day by September 2014
- Cross-play between mobile, PC, and consoles is a growing trend, with Fortnite, Genshin Impact, and Among Us as examples

- Almost 83 billion mobile games were downloaded in 2021, 64% of people who play mobile games do so on a daily basis
- Today, 4% of apps on Android Play Store are paid
- Temple Run released for \$0.99 in 2011, but switched to Free to Play after one month, increasing revenue by 10x through in-app purchases
- Popular monetization models today: (1) Paid games (one-time payment), (2) Free to play with in-game transactions, (3) Free to play with advertisements, (4) Subscriptions
- Proceeds to discuss popular mobile gaming categories, trends, and future looking information

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LO 8: Students will be able to discuss potential future trends in gaming, reflecting on emerging technologies, cultural shifts, and player expectations.

REQUIRED

The Summer of Pokemon Go

YOUTUBE | ~18 MIN

<https://www.youtube.com/watch?v=Gm-UVS7ETc>

This video discusses the history and craze behind Pokemon Go, the first augmented reality (AR) game to reach critical mass. AR is a technology that enhances the real world by overlaying computer-generated digital information onto the user's view of the physical world. We'll be discussing AR more when we discuss Extended Reality, including Virtual Reality and Mixed Reality, but I share this now because Pokemon Go is played on a smartphone, so is an extension of mobile gaming.

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REQUIRED

How Free Games Trick You Into Spending Money

YOUTUBE | ~17 MIN

https://www.youtube.com/watch?v=uvjjOtxSKdQ&ab_channel=GoingIndie

This video discusses the history of free to play and pay to win monetization strategies. Riot Games popularized free to play in modern games with League of Legends. The video includes strategies for monetizing free to play games, including the pay to win model.

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REQUIRED

Fortnite Players Get Millions in Refunds for Unwanted Purchases

ARTICLE | ~4 MIN

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1oa9cn5/cdi_proquest_miscellaneous_3143877489

The Federal Trade Commission accused Epic Games of using "dark patterns" to trick millions of Fortnite players into making unwanted purchases. A press release from the FTC (found in additional resources) cites confusing button configurations, making it easy for children to make purchases without parental consent, and Epic locking the accounts of customers who disputed charges. Epic agreed to pay a \$275 million penalty as settlement, to be distributed to customers who believe they have been injured by the practices. Epic is

prohibited from using these practices and charging customers without affirmative consent, and also from blocking customers from their accounts for disputing charges.

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REQUIRED

Mass market, mystery boxes and metric-driven design: The legacy of FarmVille

ARTICLE | ~14 MIN

<https://www.gamesindustry.biz/mass-market-mystery-boxes-and-metric-driven-design-the-legacy-of-farmville>

This article is not about portable gaming, but about the free to play model and monetization strategies that have become common in mobile games, so I thought it would be worth sharing. Highlights:

- Discusses the casual farming simulation game FarmVille, played in a Web browser (within Facebook), that became a huge success
- Produced in six weeks, had hit 1 million players within 5 days!
- Free to play
- Encouraged players to promote the game to their social network within Facebook
- Players who were playing every day didn't consider themselves gamers
- Led to the explosion of social and free-to-play games
- At one point had 80 million daily players
- To pay the bills, they introduced an in-game currency purchased with real money that was then used to buy in-game items
- Introduced daily bonuses, free items given to players for logging in each day, a common mechanic in many video games today
- Introduced "mystery boxes" (now commonly known as loot boxes) that granted players a random reward - there is a lot of scrutiny today over whether the modern implementations of loot boxes constitute gambling
- Shut down in 2020
- "showed developers, investors and most importantly players that there was a world of games that could appeal to anyone"

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REQUIRED

Free To Share Games Just Got SERIOUS

YOUTUBE | ~16 MIN

<https://www.youtube.com/watch?v=rhbBsZwiJIM>

Highlights:

- Split Fiction is an example of a relatively new idea: free to share games, where one person purchases a license and can share free access to another person (or other people) that they will play with
- Split Fiction is 2-person co-op - it requires two players and only two players to play together
- A handful of other games have been released under this model
- Sunderfolk takes the concept in a new direction, where one person buys and runs the full game and several others control their characters via a free mobile app, similar to how a group of friends might share a tabletop game or board game - one person buys it and everyone plays together (I've played this game and it's a lot of fun, and I think this is a really interesting model that could be a great way to get more people playing together)
- Likens the model to the Jackbox games

- The video points out that a game like R.E.P.O. is \$10 per person but if you have 5 friends playing together, that is essentially the same price as Sunderland

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EXPLORE FURTHER

The Game Boy Advance Chronicles, a brutally sacrificed console | A Game Boy Advance Documentary

[YOUTUBE](#) | ~75 MIN

<https://www.youtube.com/watch?v=92YpXg738ps>

This longer documentary starts by discussing some of the history we've already covered, but then goes into a discussion of the history and technology of the Game Boy Advance, the first true successor to the Nintendo Game Boy. Highlights:

- Includes discussions of some unique related technologies and accessories
- Discusses Nintendo's efforts to market the device to adults
- Also discusses Nintendo's decision to release the Nintendo DS, a new handheld in competition with the Game Boy Advance, and other competing products from other companies including the PlayStation Portable
- Despite high sales and several revisions, the Game Boy Advance was sunset in 2010

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EXPLORE FURTHER

Xbox Ally X vs Steam Deck OLED vs Lenovo Legion Go 2: BEST Gaming Handhelds of 2026

[YOUTUBE](#) | ~29 MIN

<https://www.youtube.com/watch?v=Zgjn3jsExco>

Provides a good overview of the features, capabilities, and accessories of several modern handheld gaming devices.

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EXPLORE FURTHER

The Indie Game Development Podcast, May 8, 2024: How Free Games Trick You Into Spending Money

[PODCAST](#) | ~22 MIN

<https://sasquatchbstudios.podbean.com/e/how-free-games-trick-you-into-spending-money/>

An indie game developer calls out game developers for using psychological profiles to take advantage of players with addictions.

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EXPLORE FURTHER

FTC Finalizes Order Requiring Fortnite maker Epic Games to Pay \$245 Million for Tricking Users into Making Unwanted Charges

PRESS RELEASE

<https://www.ftc.gov/news-events/news/press-releases/2023/03/ftc-finalizes-order-requiring-fortnite-maker-epic-games-pay-245-million-tricking-users-making>

FTC press release providing more information about the Fortnite settlement.

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EXPLORE FURTHER

Monetisation In Video Games

ARTICLE | ~22 MIN

<https://gameopedia.com/blogs/monetisation-in-video-games>

Provides an overview of various monetization models used historically and today.

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EXPLORE FURTHER

Where the Whales Live: The Pyramid Model of F2P Design

VIDEO | ~39 MIN

<https://www.gdcvault.com/play/1019671/Where-the-Whales-Live-The>

A talk from a 2013 Game Developers Conference that describes how developers can get more money out of the free to play model. Kind of a dark talk, but provides insight into the design decisions that go into free to play games.

EXPLORE FURTHER

New York Targets Valve's Loot Boxes as Illegal Gambling

ARTICLE

<https://www.joneswalker.com/en/insights/blogs/perspectives/new-york-targets-valves-loot-boxes-as-illegal-gambling.html?id=102ml3c>

This article examines a groundbreaking legal challenge brought by the New York Attorney General against Valve Corporation, alleging that the company's randomized loot boxes function as an unauthorized gambling operation. While past private lawsuits failed because courts viewed in-game items as lacking measurable monetary worth, this new litigation argues that Valve's virtual "skins" are convertible to real-world cash through both official and third-party marketplaces. The author highlights how New York's specific penal laws and Valve's alleged facilitation of secondary markets create a unique threat to the gaming industry's traditional monetization models. Ultimately, the article serves to warn developers that the legal immunity surrounding digital prizes is shifting, especially regarding systems that utilize psychological triggers to encourage spending among minor children.

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EXPLORE FURTHER

Valve moves to dismiss Counter-Strike gambling lawsuit in New York

ARTICLE

<https://www.courthousenews.com/valve-moves-to-dismiss-counter-strike-gambling-lawsuit-in-new-york/>

This article provides an update on the New York attorney general's lawsuit against the gaming giant Valve. Valve is seeking to dismiss the lawsuit. In its defense, Valve attempts to reframe these digital items as modern collectibles, comparing them to baseball cards or mystery toy packs rather than illegal wagers. Ultimately, the article highlights the potential industry-wide implications of the case, as a ruling against Valve could fundamentally change how digital goods are sold and regulated across the United States.

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EXPLORE FURTHER

Masters of Doom: How Two Guys Created an Empire and Transformed Pop Culture

BOOK

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1i25tbk/alma991045717659708681

A book by David Kushner about id Software and its influence on popular culture, focusing on co-founders John Carmack and John Romero. I meant to include it last week, sorry.

EXPLORE FURTHER

Did This Game Kill Three People?

YOUTUBE | ~37 MIN

https://www.youtube.com/watch?v=LgZoCyd_cSk

This is about an arcade game that may have been involved in the deaths of 3 people, or maybe only one person, but even that is questionable. I meant to share this a couple of weeks ago, but when I realized my omission I figured I would share it even though it's off-topic for this week.

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